

Experimental commute calibration - reviewed report

Four corrected report pages; 17 selected-case diagnostic pages copied unchanged from frozen export

Numerical result

Selected weighted loss 381.0485741441105; 12 scored moments, 8 free coordinates, and a separate completed-fertility normalization to 2.1.
23 objective cases scored; 1 additional case timed out inside a native solve; 17 planned cases unrun.
143 native stationary solves completed; 1 began but did not complete. Export met the one-hour deadline.

Target choice pending clarification

The frozen experiment scored a first-birth room response target of 1.465.
This was our interpretation of the author's phrase 'the new one'; the earlier household target is 1.025.
The author has questioned that interpretation. No target edit, rescore, or numerical rerun is implied.

Search-bound review failure

Actual frozen annual discount-factor (beta) search bound: [0.94, 0.9995]. Requested upper cap: 0.99.
This is a preflight/review error. All 23 scored betas were ≤ 0.9762497404726131 , including the selected beta 0.9762497404726131; the contract itself did not enforce 0.99.

Measurement and specification limits

First-birth housing uses an unmatched stationary model proxy for the PSID panel response.
The model bequest row measures gross positive estates, including childless households; the accepted SCF target is child-directed. The adopted 16/20 birth-entry queue is unimplemented.
PAYGO tax 0.08751017424959717 is experimental, not adopted.

Plot note: owner demand concentrates at the 10-room maximum in the source figure; cause unproven.
Income-state labels in the unchanged source figures remain crowded.

Complete target fit

Exact frozen 13-row objective; positive weights retained on all 12 scored rows

Moment	Target	Model	Gap	Weight	Loss contribution
initial_normalization	2.1	2.10000052598	5.25978920418e-07		
cps_childlessness	0.198278751007	0.168814364906	-0.0294643861004	35532.3042455	30.8473716461
cps_exactly_one	0.21365532522	0.239169182287	0.0255138570669	26952.8208243	17.5451247556
nchs_mean_age	25.976263861	26.0789058849	0.102642023919	139.828067845	1.47314253892
nchs_share30	0.249278013041	0.228389763504	-0.0208882495371	13866.0654347	6.05002737072
wealth_earnings	6.92658379107	7.26789385757	0.341310066494	7.59509847253	0.884772475838
bequest_wealth	0.00729102347262	0.00672738593208	-0.00056363754054	5165289.2562	1.64094667926
old_dispersion	3.51593508652	2.86742702527	-0.648508061251	10.6163615315	4.46484572835
mean_rooms	5.60788596007	6.47376746165	0.865881501578	128.020702052	95.9836205509
ownership_30_55	0.676260416854	0.40575669235	-0.270503724504	2339.36237247	171.17644338
first_birth_rooms	1.465	0.978604331599	-0.486395668401	137.5652749	32.5452953925
family_rooms	0.385099649693	0.282272006996	-0.102827642696	280.528083702	2.96617045445
recent_parent_ownership	0.127608363567	0.151520930384	0.0239125668172	27055.8229575	15.4708131711

Gap = model - target. The fertility-normalization row has no objective weight.

Free parameters and normalized inputs

Actual frozen search bounds shown without rounding; near bound uses 1% of the stated interval

Annual beta upper bound 0.9995 was erroneous; requested cap 0.99. Selected beta is below both.

Parameter	Estimate	Lower	Upper	Near bound	Status
H0	8.9672156595	0.2	80	False	experimental free coordinate
beta_annual	0.976249740473	0.94	0.9995	False	experimental free coordinate
chi	0.887984968017	0.1	5	False	experimental free coordinate
first_birth_fixed_cost	0.506777288217	0	8	False	experimental free coordinate
h_P	1.89004766001	0.1	2.3	False	experimental free coordinate
kappa_fert	0.209275857605	0.02	50	True	experimental free coordinate
kappa_fert_continuation	0.482068766683	0.02	50	True	experimental free coordinate
theta0	0.0881692077743	0	8	False	experimental free coordinate
theta1	0.008193084127				externally fixed B15
psi_child	0.12165036564				normalized to 2.1
payroll_tax	0.0875101742496				experimental PAYGO rate
pension_period	1.00043263566				endogenous balanced PAYGO

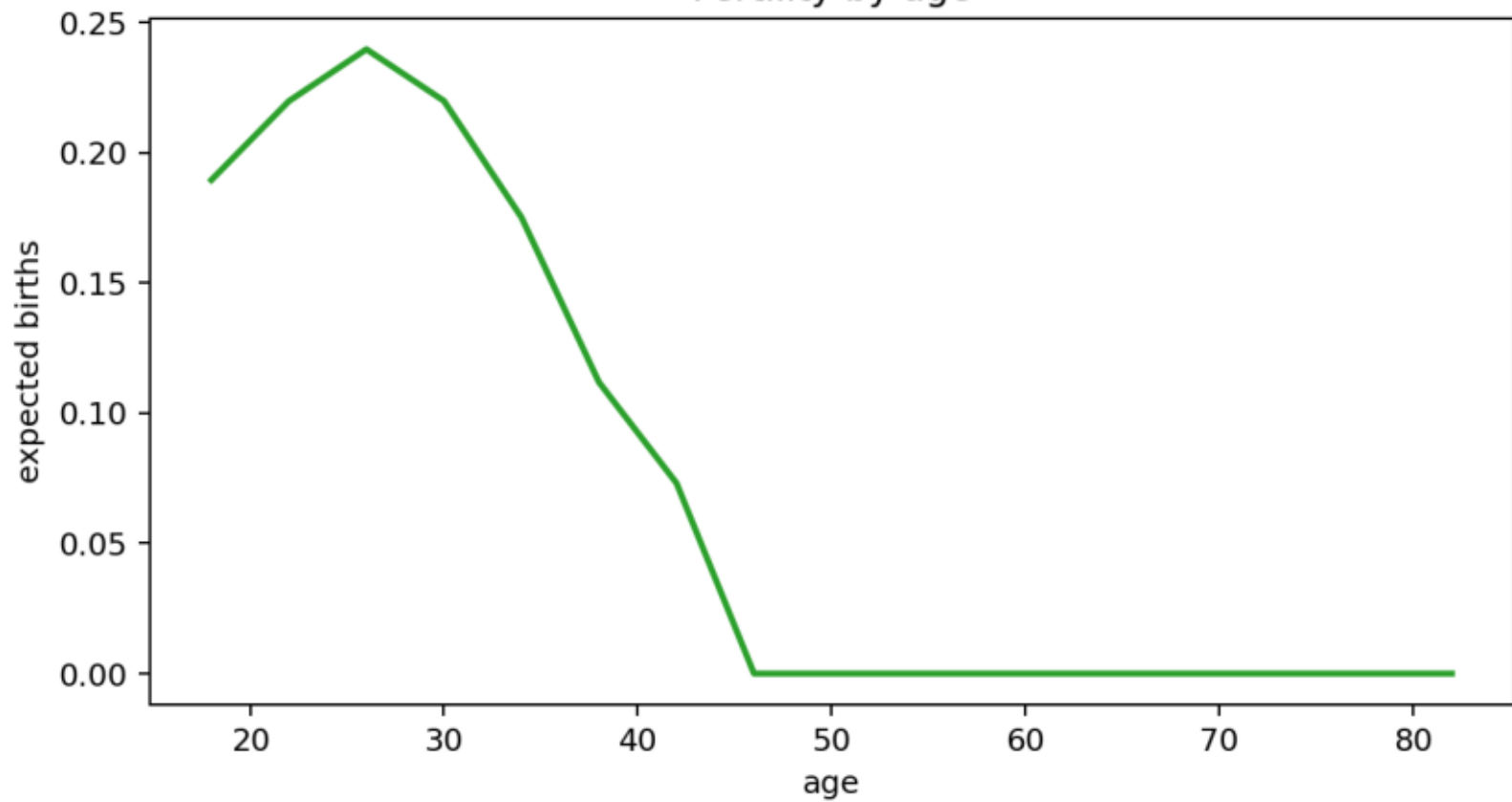
Retained settings and annual-to-period inputs

Four-year depreciation is compounded; property tax follows the source's four-year linear convention

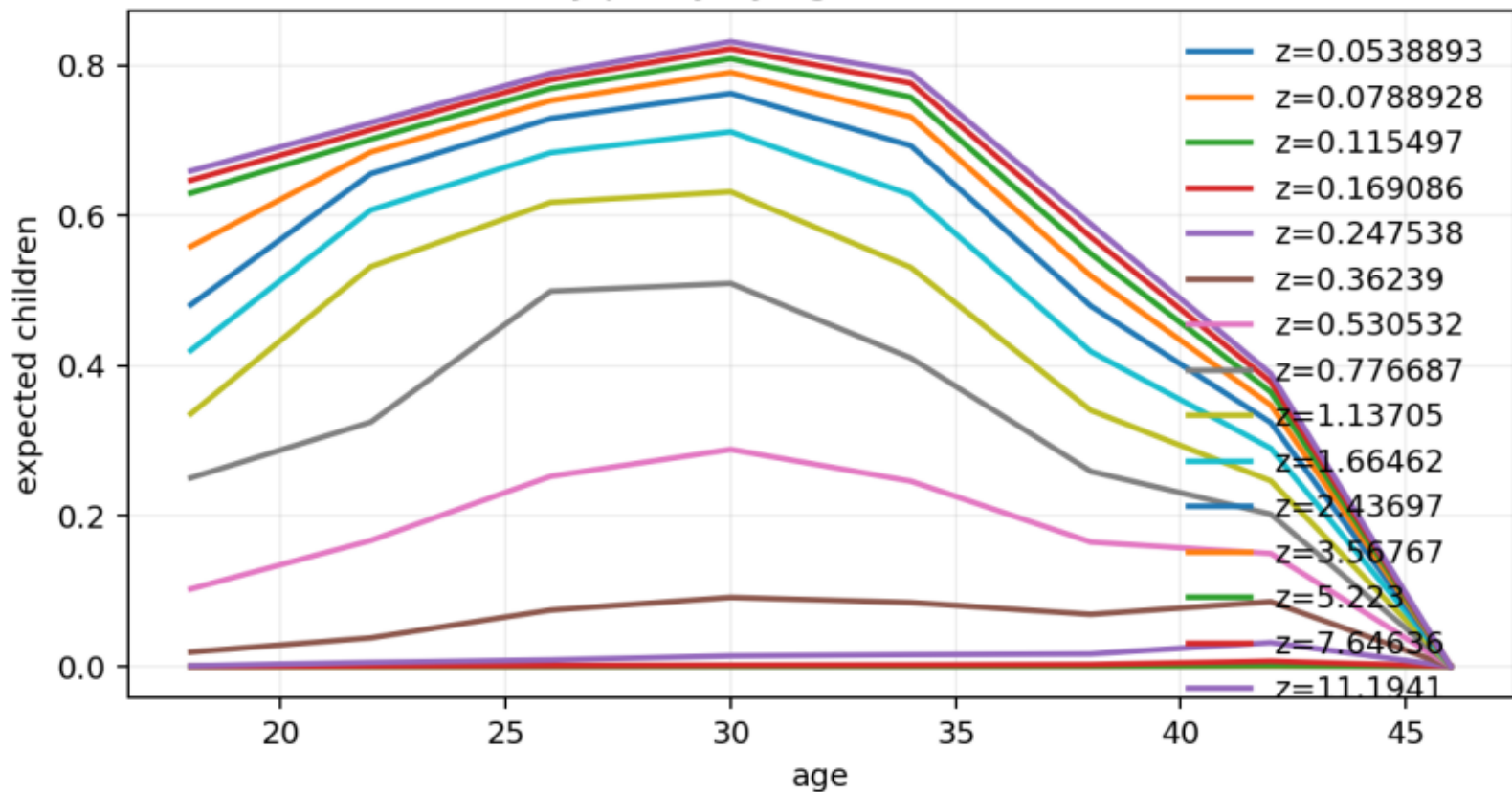
Parameter	Estimate	Lower	Upper	Near bound	Status
housing_supply_elasticity	0.63				retained external setting
tenure_choice_kappa	0.005				retained external setting
alpha_cons	0.733				retained external setting
sigma	2				retained external setting
selling_cost	0.06				retained external setting
financed_share	0.8				retained external setting
annual_depreciation	0.0141614371838				author adopted input
period_depreciation	0.0554537907933				four-year compounded
annual_property_tax	0.0105983607739				author adopted input
period_property_tax	0.0423934430955				four-year linear source convention
income_process	15				retained B15 persistent-state count
entrant_conversion_factor	0.5				retained stationary law

fertility_by_age

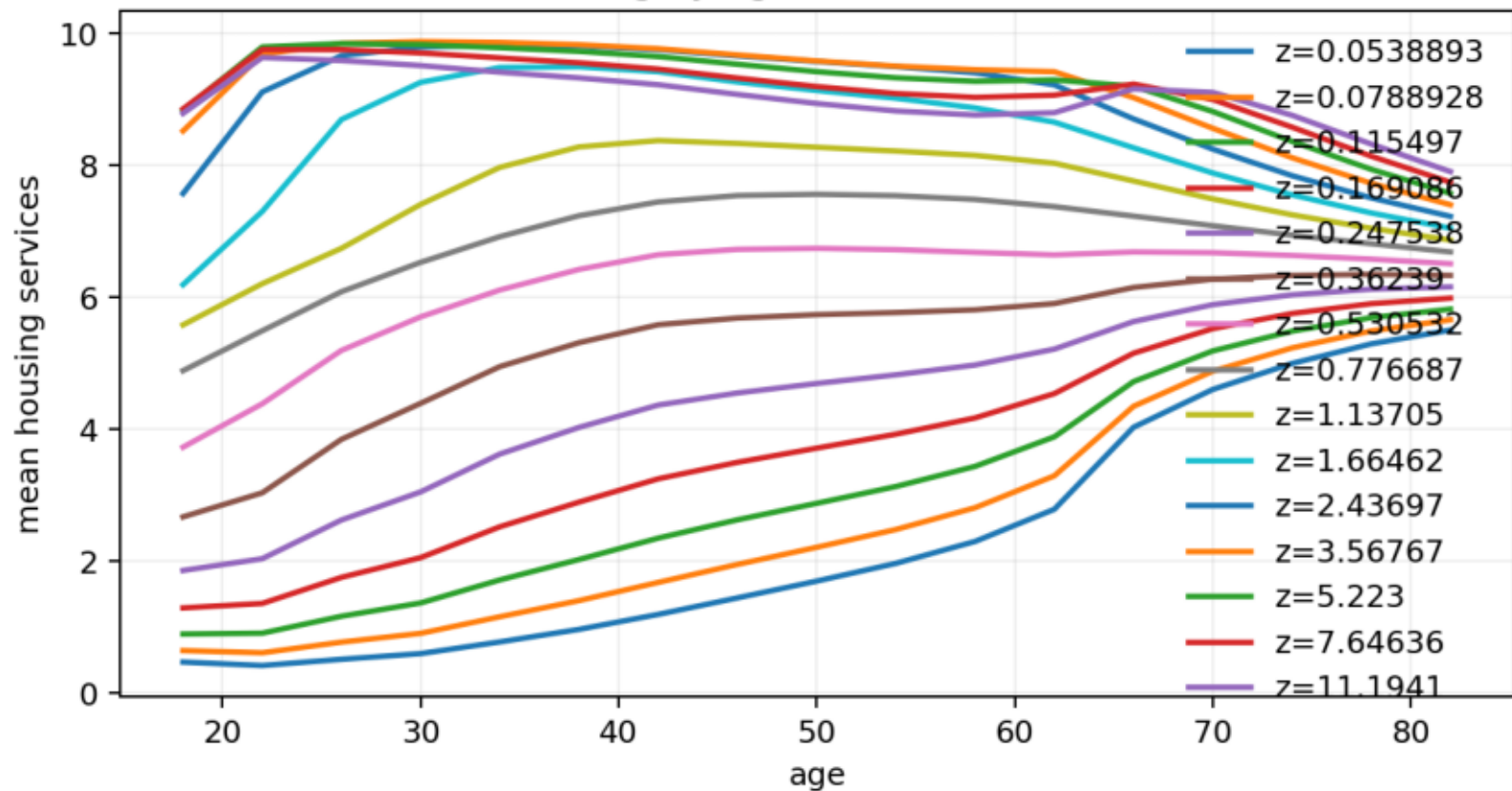
Fertility by age

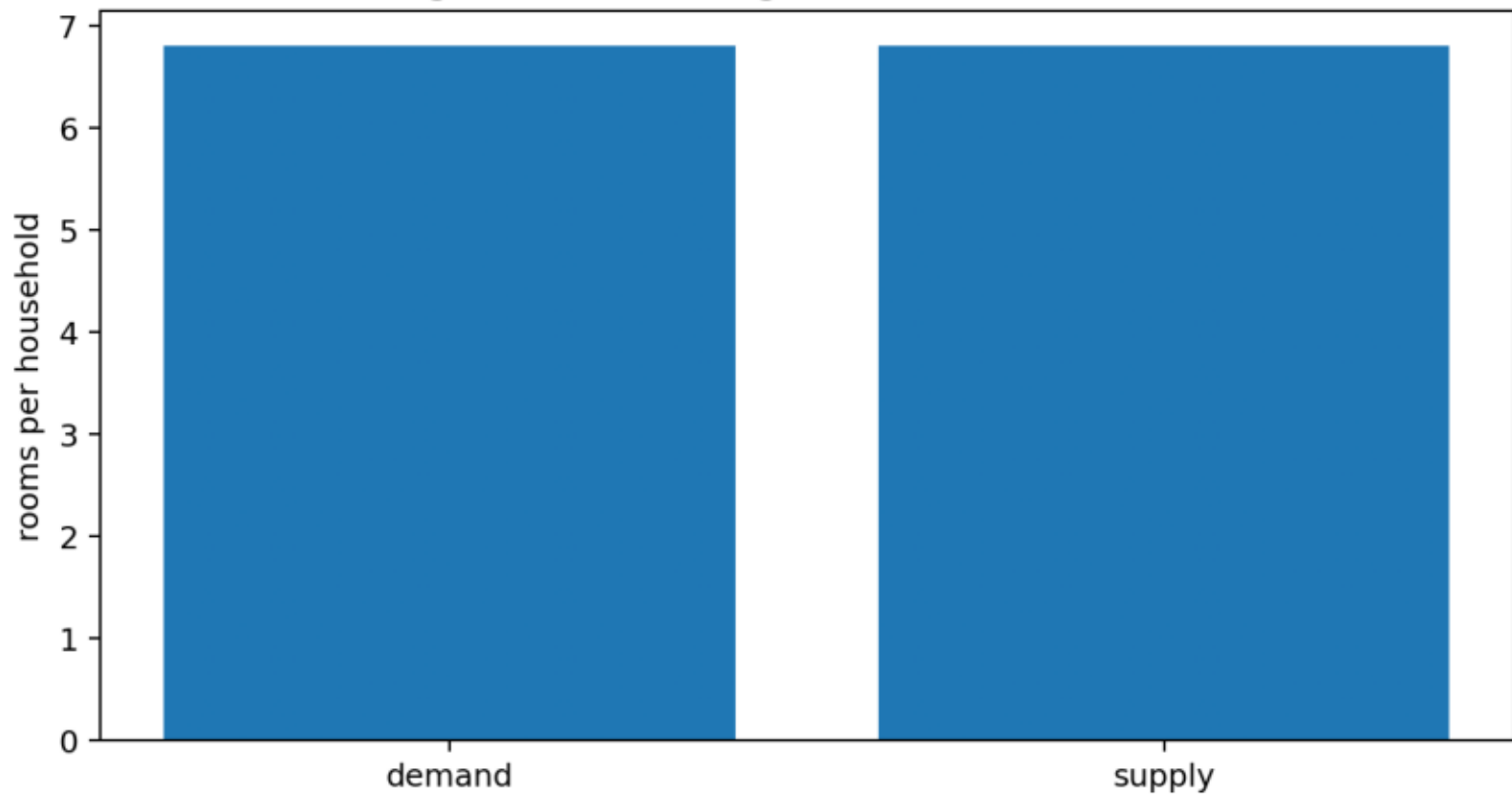


Fertility policy by age and income state

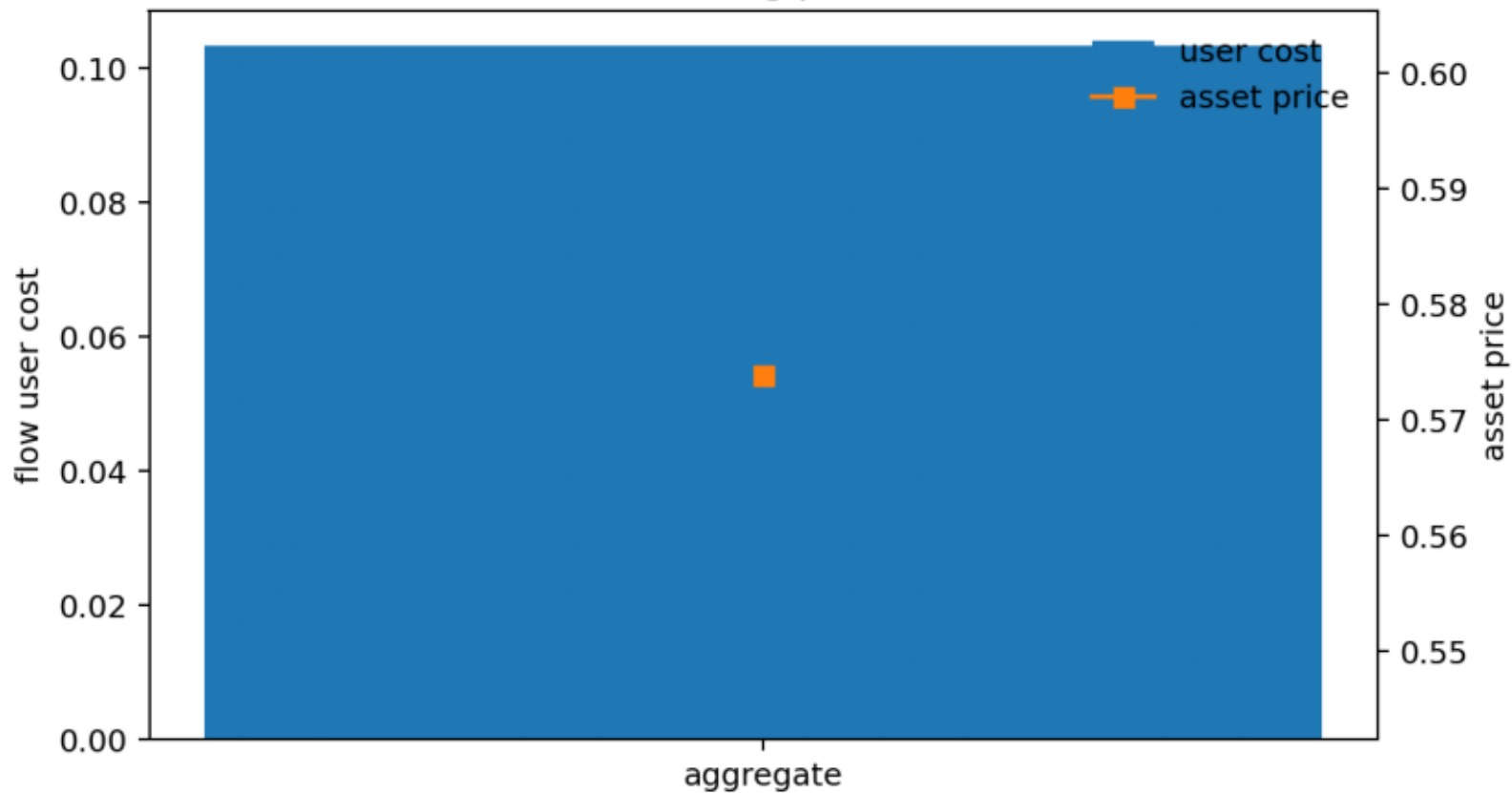


Housing by age and income state

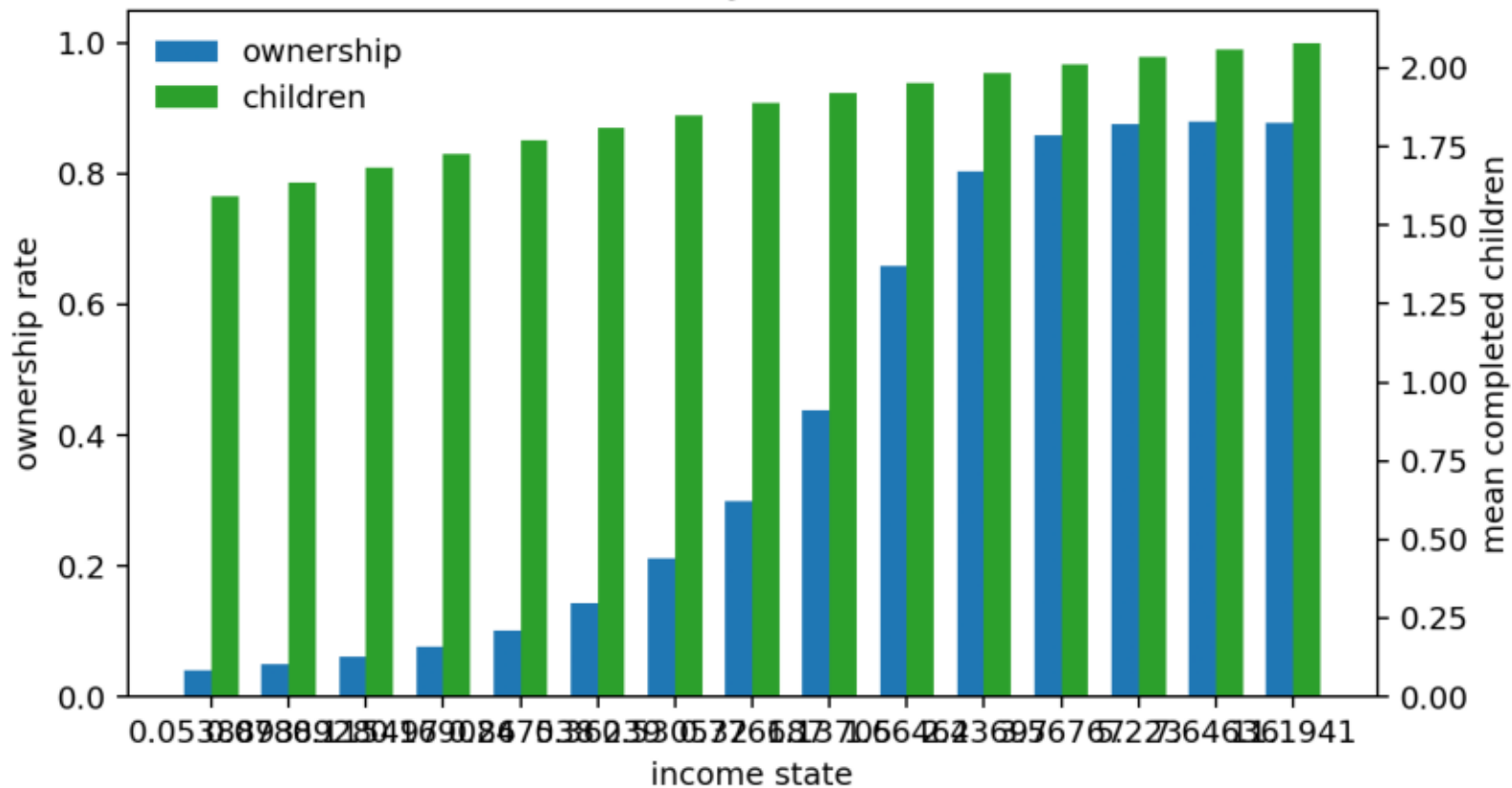


Housing-services clearing, max rel. residual= $1.81\text{e-}06$ 

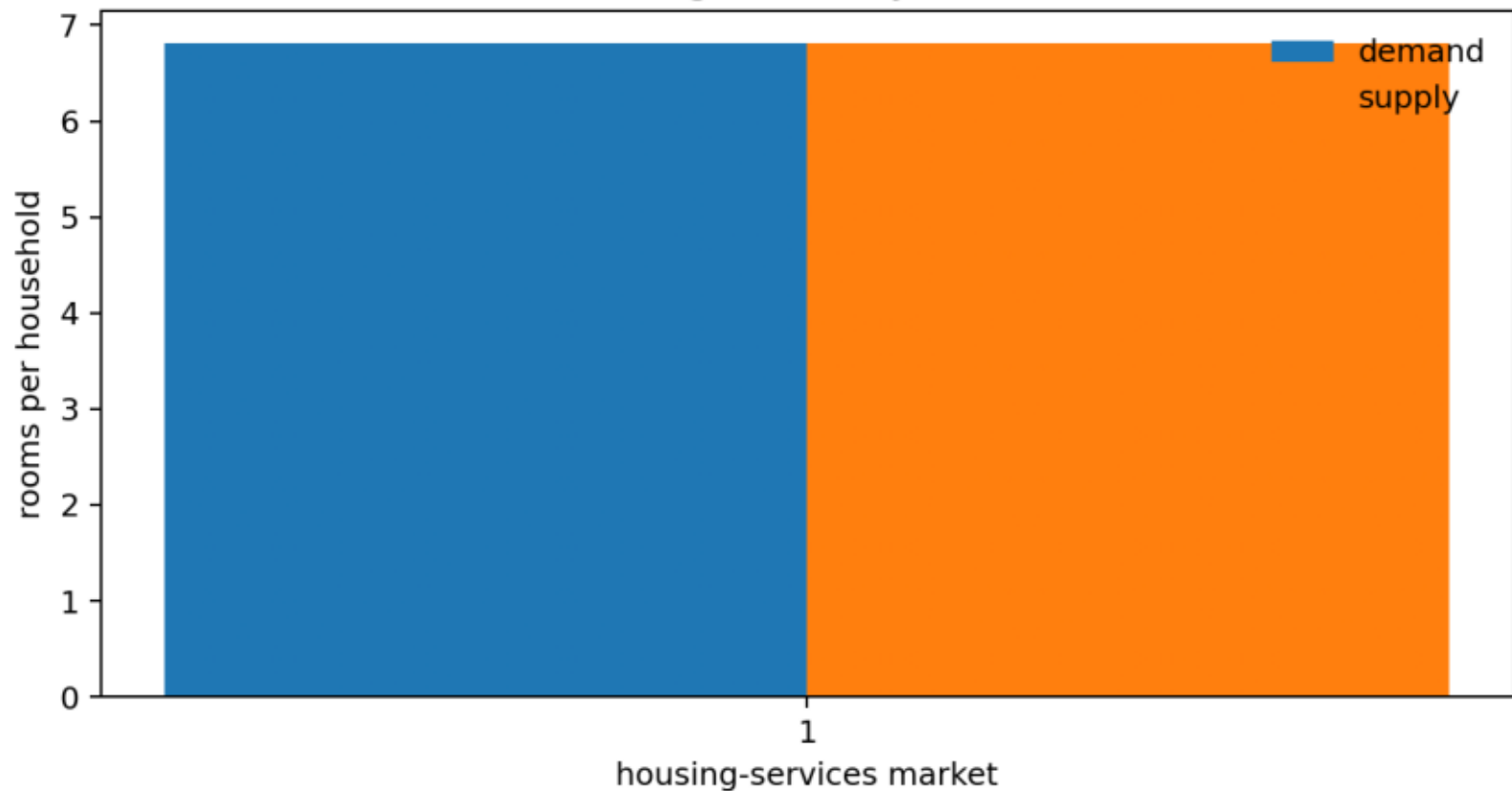
Housing price

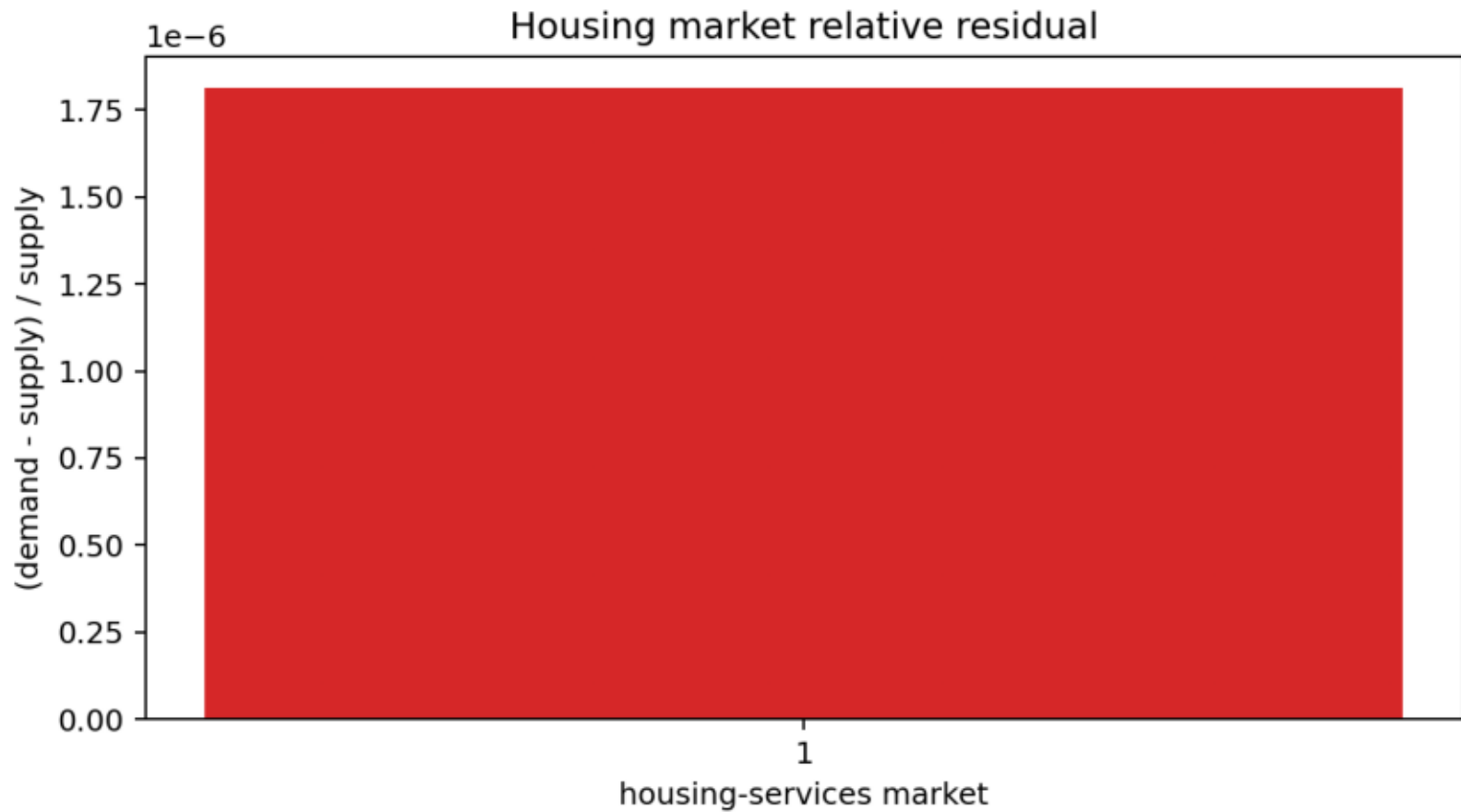


Outcomes by income state



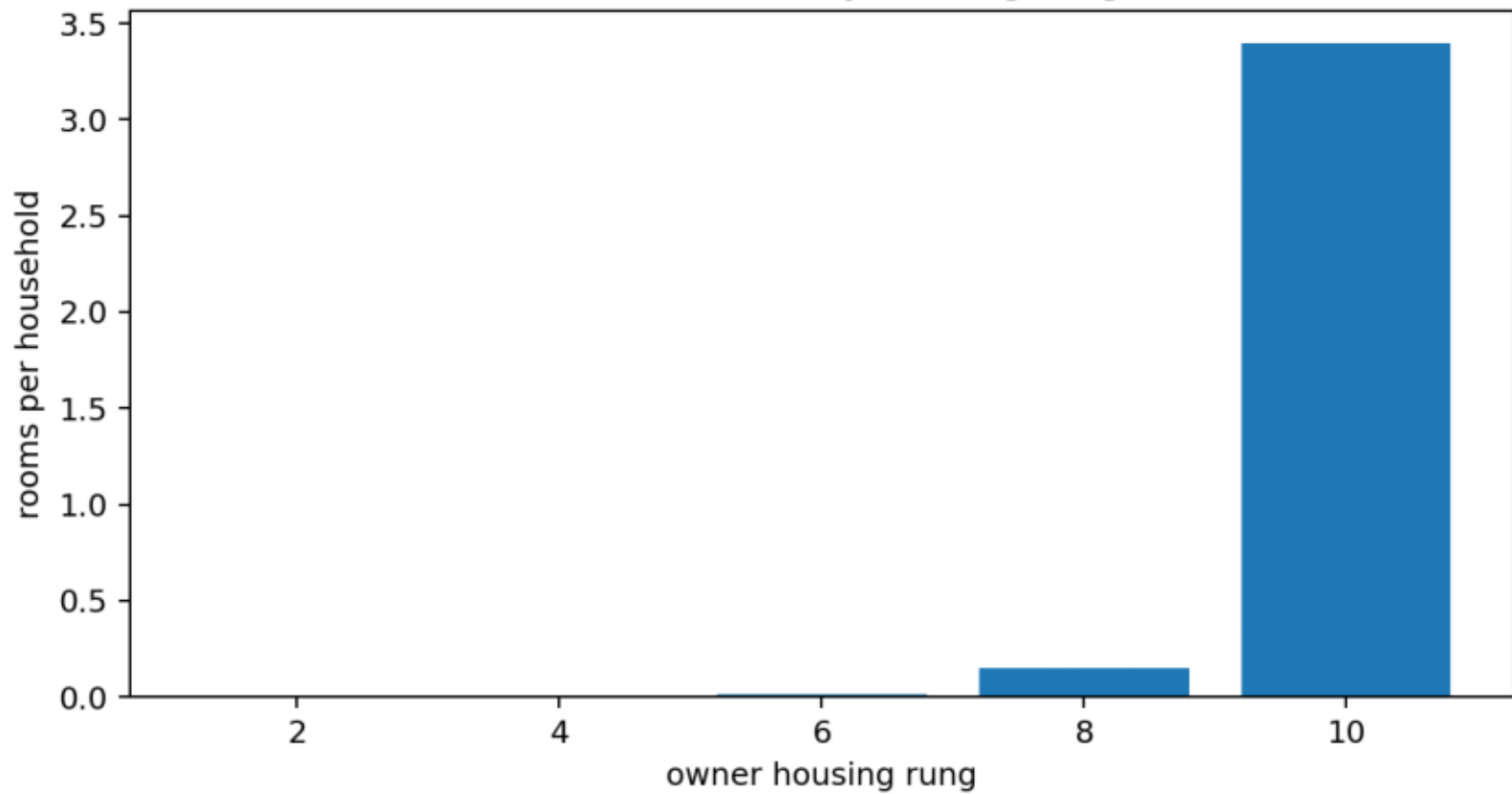
Housing market by market



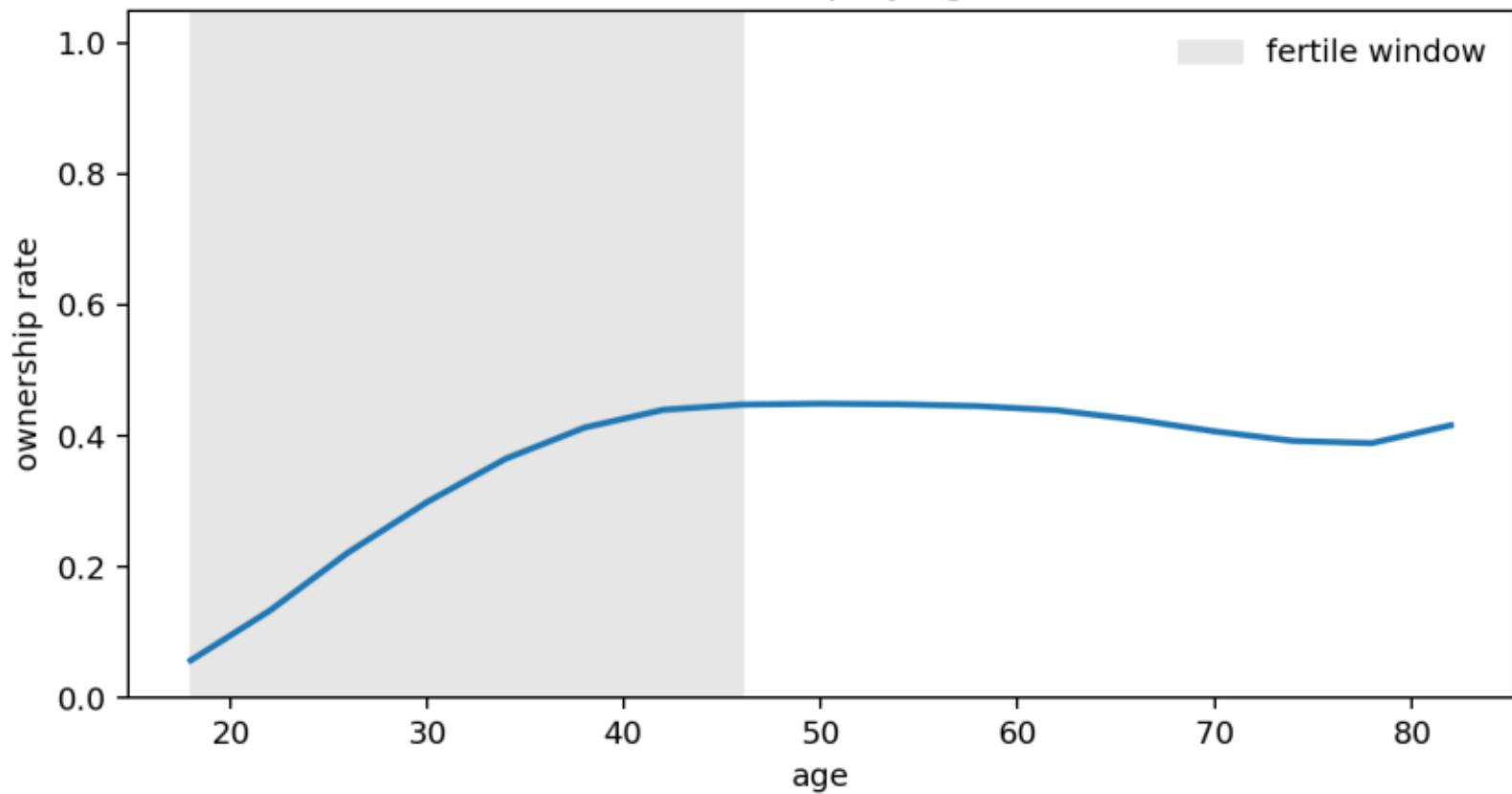


owner_rungs

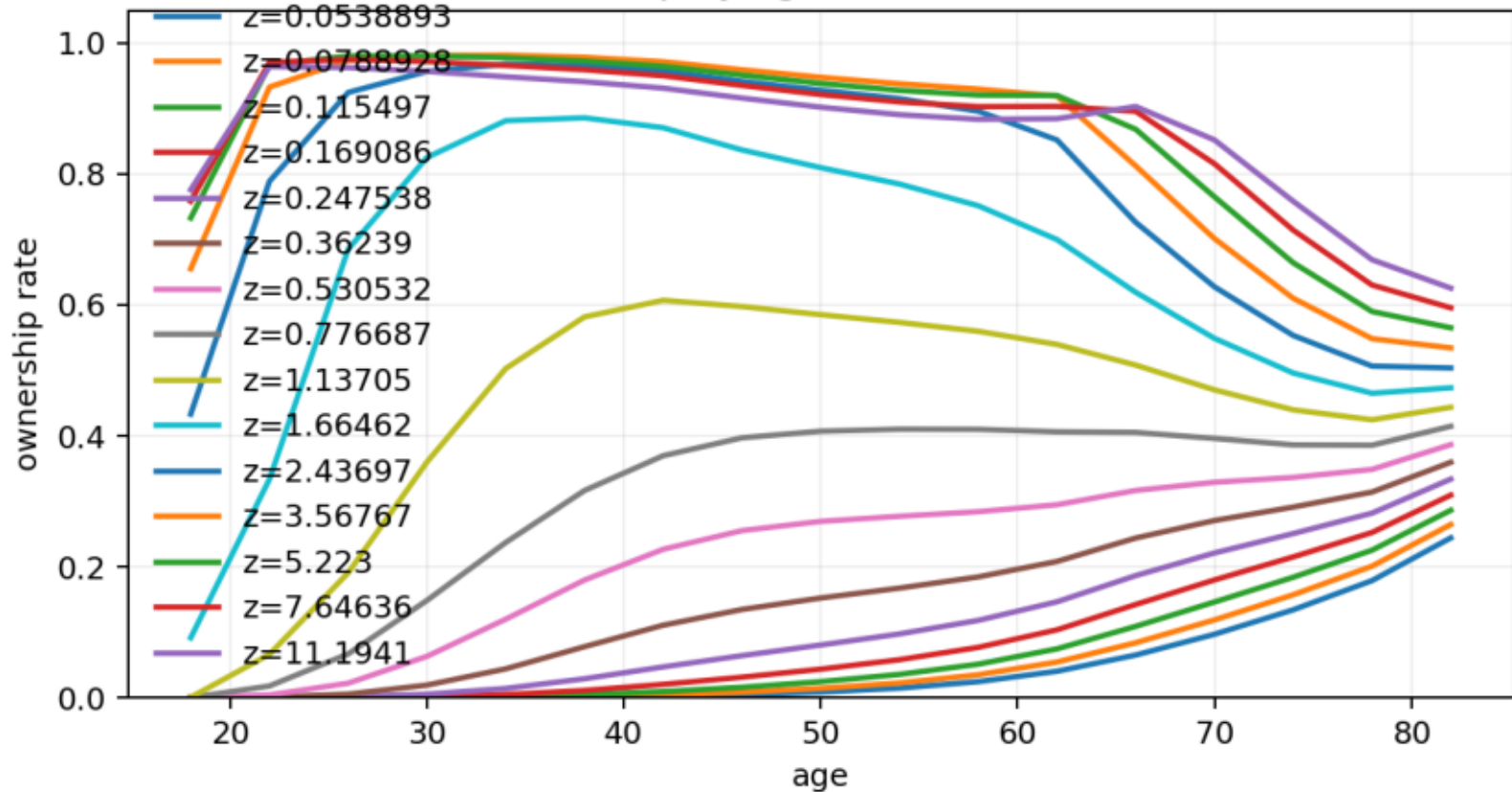
Owner demand by housing rung



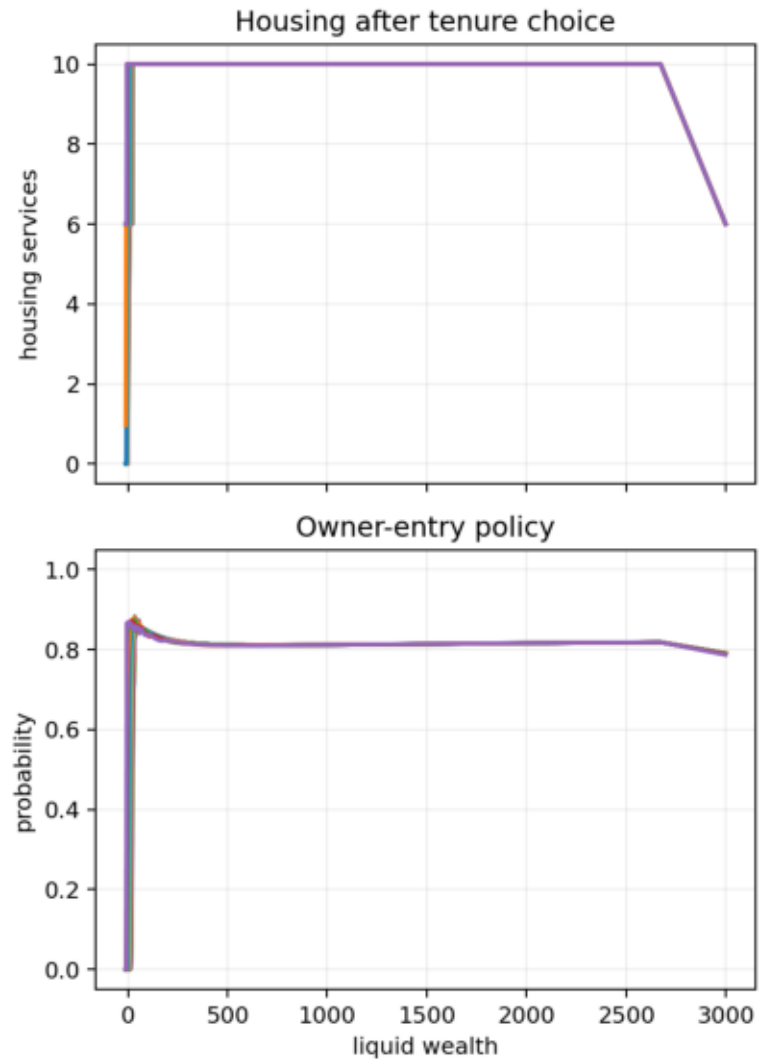
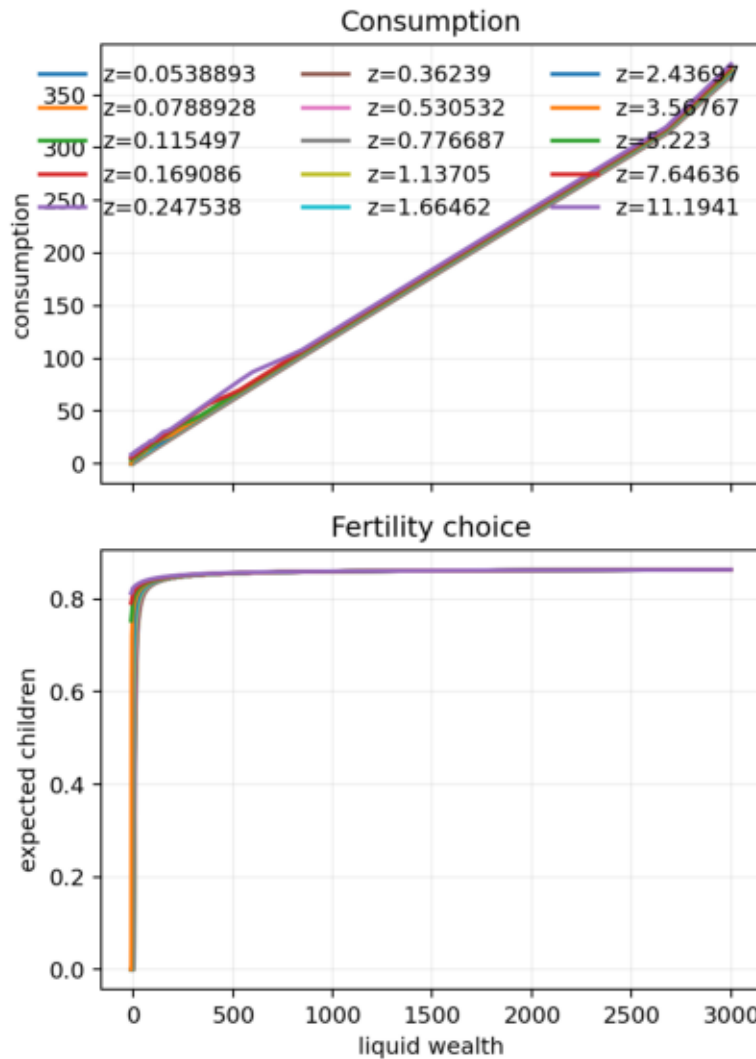
Ownership by age



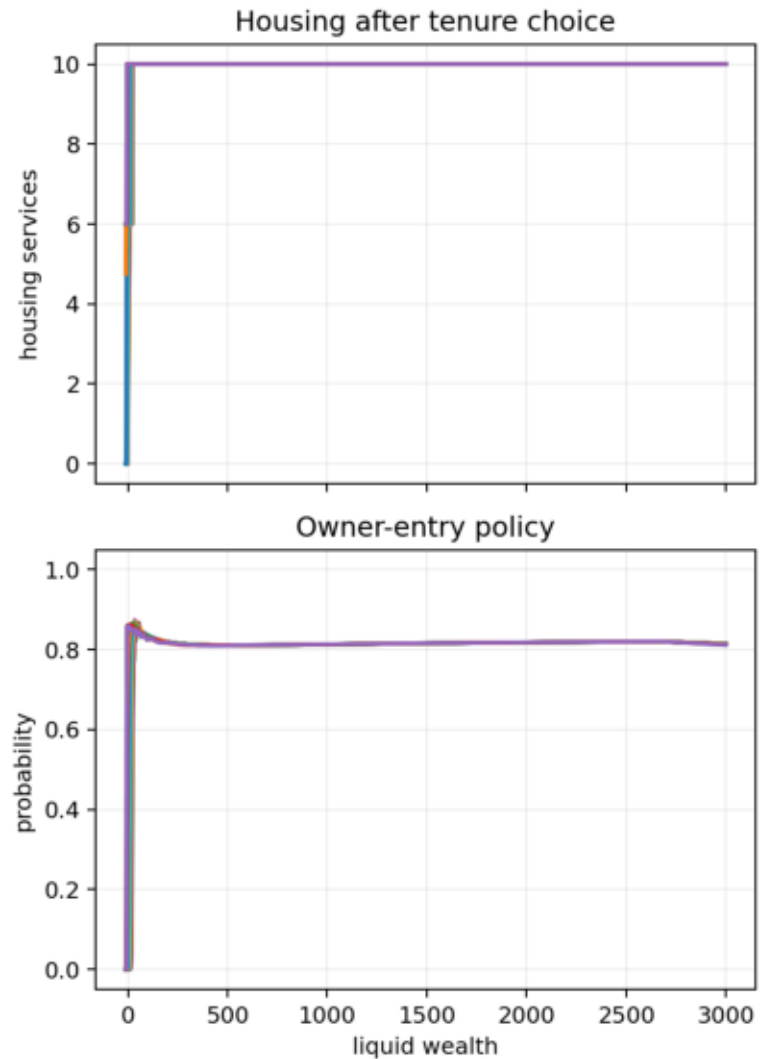
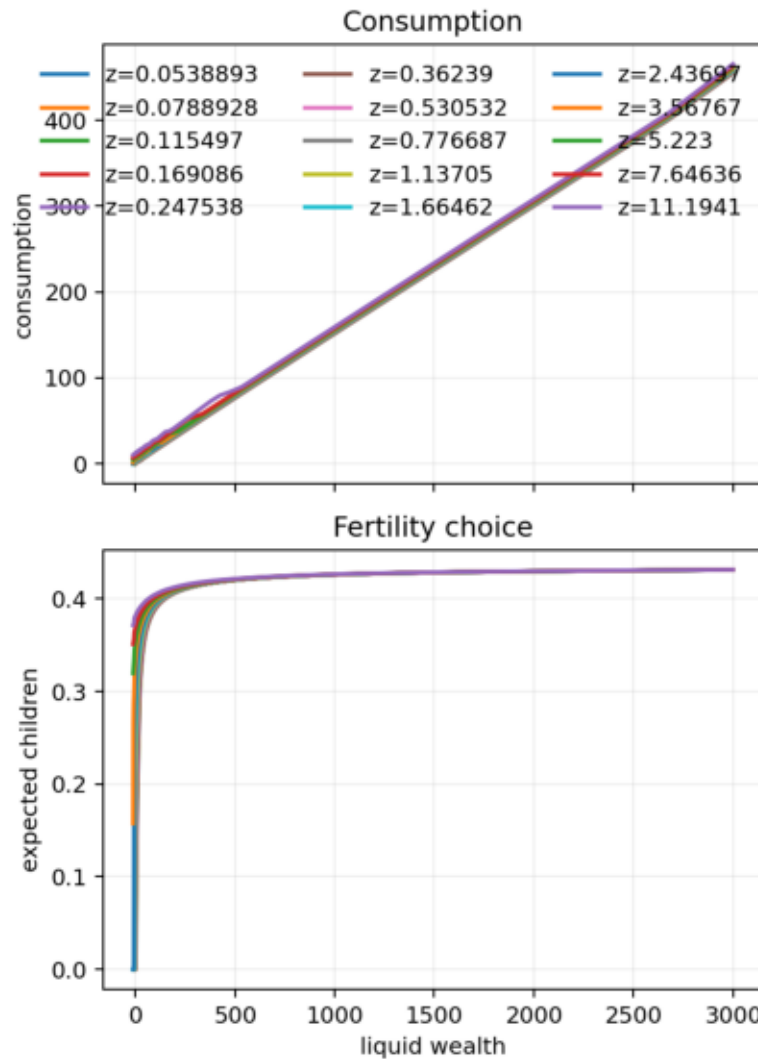
Ownership by age and income state



policy_childless_renter_age30
Childless renter policies at age 30

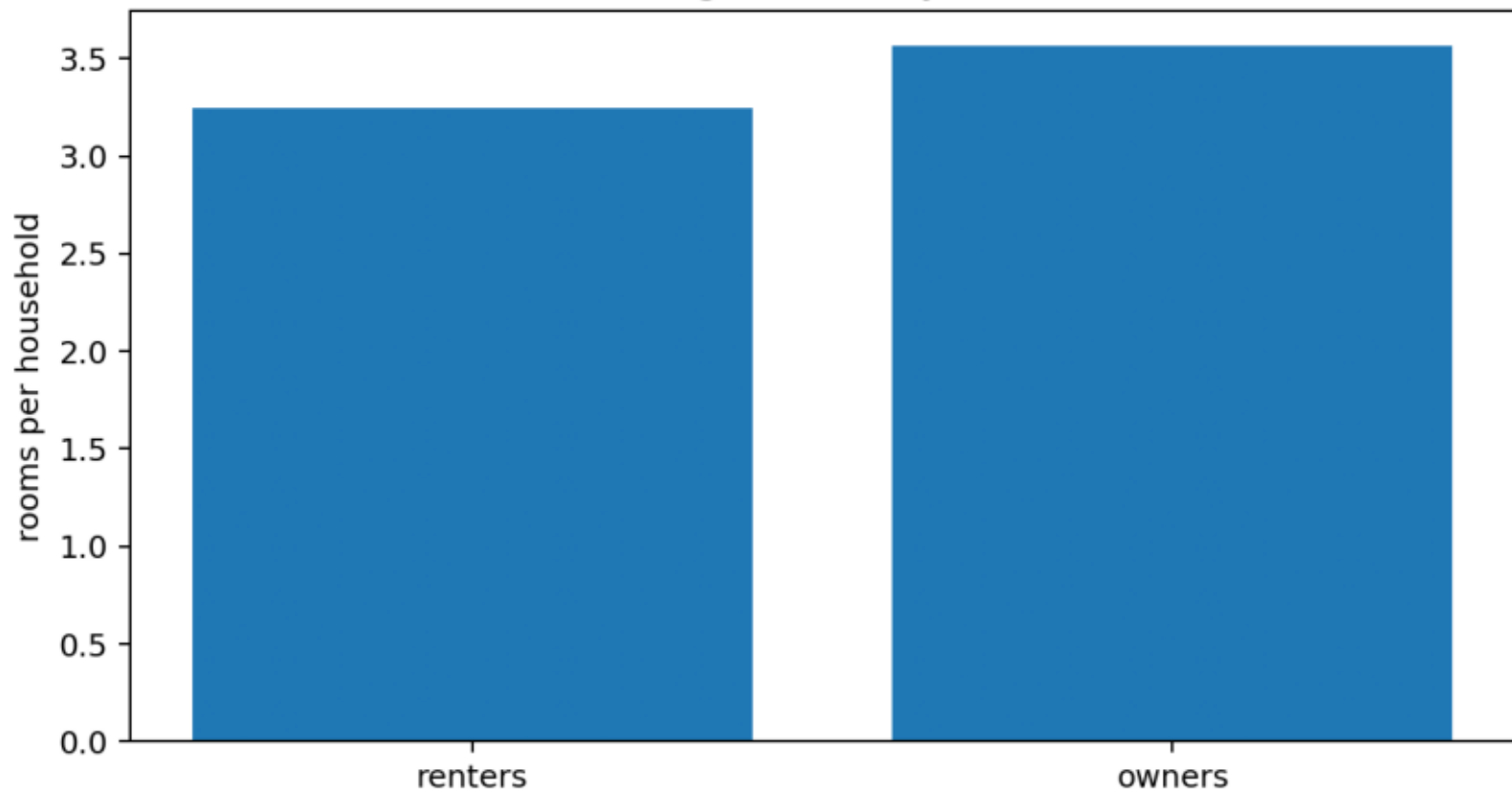


policy_childless_renter_age42
Childless renter policies at age 42

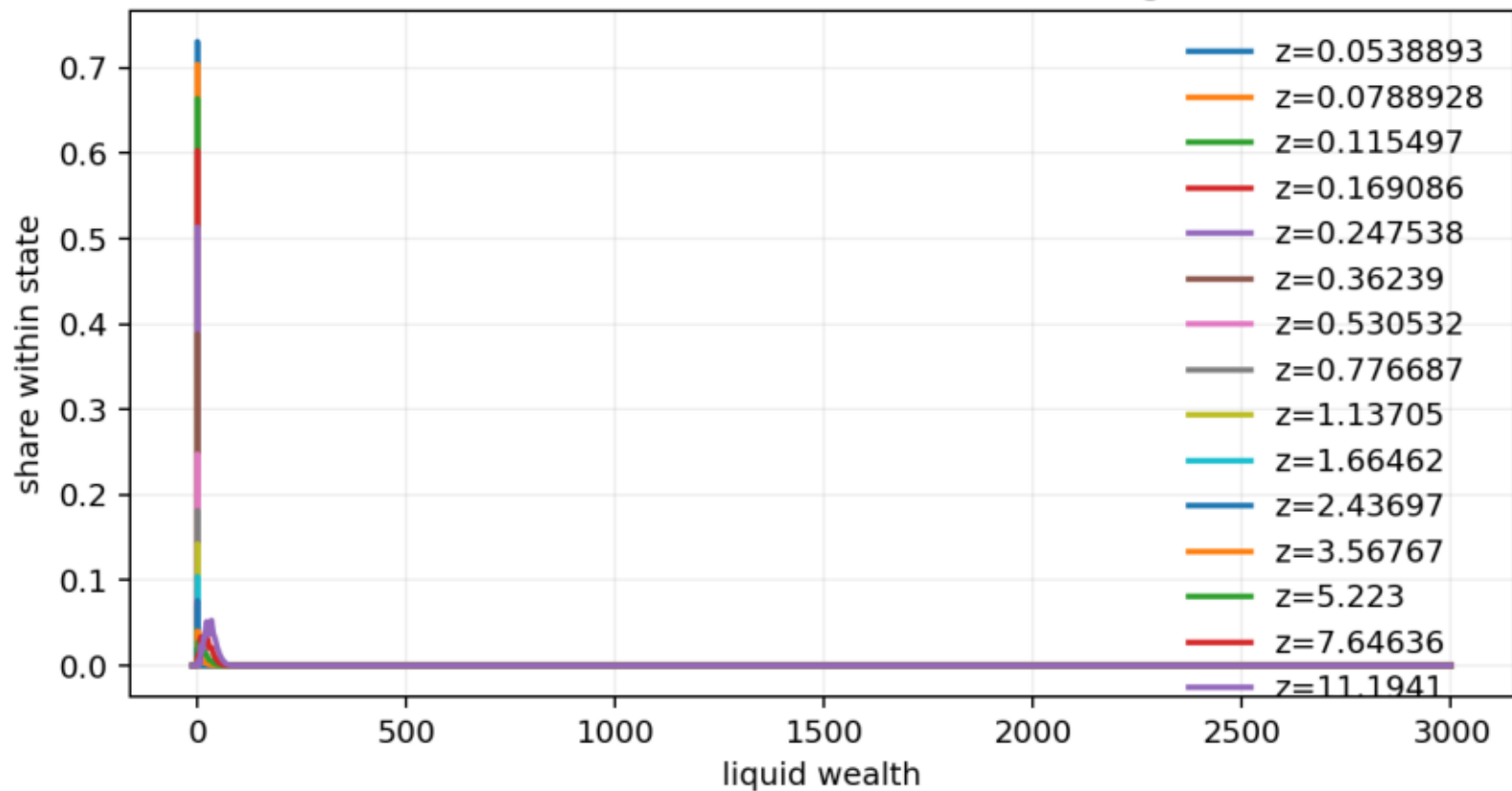


tenure_services

Housing services by tenure



Childless-renter wealth distribution at age 30



Childless-renter wealth distribution at age 42

